

controllers, processor resources, etc. It loads drivers into memory for the detected hardware. After hardware initialisation, it creates an area in the RAM called a RAM-disk, to store the necessary OS components. The Live CD then loads the required OS components into the RAM-disk, and then loads up the services and applications needed for Live CD functionality.

Frankly speaking, the process is more complex than I've explained. This is because with a hard drive installation, at boot time, the location of the root filesystem is fixed; it is passed to the kernel using the `root=/dev/...` parameter. However, in a Live CD, the location of the root device is not fixed, since the user might have multiple CD-ROM drives. Also, to fit on a CD, the filesystem is compressed using *squashfs*, and it requires proper modules to mount it with transparent decompression.

Although it is true that CDs are still slower than hard disks, normal users generally don't care about a few milliseconds' delay. And with the advances in CD-ROM speed, the future seems promising.

Spin your own Live CD

So why not spin your own Live CD? I'm going to discuss the creation of a personal Ubuntu Live CD (from a pre-installed copy of Ubuntu) using a GUI tool called UCK (Ubuntu Customisation Kit). UCK helps users customise official Ubuntu Live CDs to their needs. Users can add any package to the live system—any applications, games, etc. To install UCK, open a terminal window, and run the following code:

```
sudo add-apt-repository ppa:uck-team/uck-stable
sudo apt-get update
sudo apt-get install uck
```

Make sure your UCK version is 2.0.10 or higher, because with older versions you cannot customise recent versions of Ubuntu like Karmic or Lucid. The UCK start-up screen shows its version—see Figure 1. You need the ISO image of the Ubuntu version you want to customise; either obtain it from the Ubuntu website, or you can create one from the installation CD: insert the CD in the CD drive, and issue one of the following commands in a terminal window to create the ISO image in your home folder:

```
dd if=/dev/dvd of=my_dvd.iso (for dvd)
dd if=/dev/cdrom of=my_cd.iso (for cdrom)
```



Note: Some systems have `/dev/sr0` as the device name for the optical disk drive, and do not mount the CD by default.

It's better if you are using the same Ubuntu version which you intend to customise. If not, do these changes:

1. In a terminal, run `gedit /etc/apt/sources.list`.



Figure 1: UCK startup screenshot

2. Change the name of the Ubuntu version to the one you are customising. For example, change Lucid to Karmic. Don't forget to reverse these changes after you've finished customising your Live CD!

Now let's move on to using UCK. Launch *Applications > Ubuntu Customisation Kit*. Choose the language packages to install; just click 'OK' if you do not want an ISO with custom languages. Do the same thing for the choice of language used during the boot-up process. Choose the desktop environment. (Make sure that your ISO image has that environment; choose GNOME for Ubuntu, and KDE for Kubuntu.)

Select the Ubuntu ISO file. Keep Windows-related files if you want to use the CD under Windows.

When the UCK package manager opens, customise the list of packages you want (remove and add packages as desired). Select *Continue Building*, and your ISO image will be saved in `/tmp/remaster-new-files/livcd.iso`. Burn the images to a CD or DVD using your favourite CD/DVD burning application.

Live CDs are a wonder-wall for me, as they enable me to work without worrying about security and storage space. They enable me to play with new Linux distros without installation. They even let me recover a crashed system! In my opinion, it is very handy for everyone to have a Live CD in their pocket. **END**

Bibliography

- [1] http://en.wikipedia.org/wiki/Live_CD
- [2] Latest UCK version: http://sourceforge.net/projects/uck/files/uck/2.4.3/uck_2.4.3-0ubuntu1_all.deb/download

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